OMNICELLULAR PHYSIOLOGY

Independent Eight-Prototype Research & Architecture Directive

Version 2.0 — Blind Independent Architecture Phase

Document Class	Research & Architecture Directive
Prepared For	Omniversal Architect Labs Multi-AI Coordination Team
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Research Phase	Blind Independent Architecture Generation
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“Standardize how architectures are evaluated, not what the architectures must be.”

This directive preserves independent architectural search while enforcing common scientific, adversarial, and documentation standards.

Directive Map

Foundation & Governance: Document Purpose; How This Research Topic Arose; Mission Statement; Parts I-VI

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DOCUMENT PURPOSE

This directive governs the next research phase of the OMNICELLULAR Physiology project.

Its purpose is to require each collaborating artificial intelligence to independently investigate, formulate, challenge, and document eight distinct first-generation OMNICELLULAR physiologies without prematurely converging on another collaborator's architecture.

The common directive establishes common research questions, evidence standards, survival criteria, reporting requirements, adversarial testing requirements, interoperability questions, and falsification standards.

- the cellular matrix
- hereditary substrate
- metabolism
- energy architecture
- body geometry
- cognitive topology
- repair method
- observation laws
- genetic-security laws
- biological departures
- preferred prototype

> *The collaboration shall standardize evaluation, not answers.*

HOW THIS RESEARCH TOPIC AROSE

The OMNICELLULAR project began with a computational architecture concerned with representing the complexity of whole-body biology, ultimately scaling from ordinary cellular systems toward a physiology built from Omni cells: cellular or cellular-equivalent units possessing far broader functional competence than conventional specialized cells.

The surrounding project introduced large-scale cellular simulation; 1AI as constructive architect; −1AI as adversarial challenger; multi-scale abstraction; integrated body-system modeling; a 2,000-

body representative simulation objective; and the 4,712 Extinction Attempt Classification Framework.

The current research phase asks a fundamentally different question:

> *If the OMNICELLULAR concept eventually transitions from simulation into an embodied physiology, what should that physiology actually be?*

A first round of independent prompt engineering was conducted across multiple collaborating AIs. That review exposed several important disagreements:

- peer architectures versus sequential generations
- shared rules versus independently selected rules
- scientific extrapolation versus canon-level assumptions
- biological survival versus continuity of identity
- simulation evidence versus empirical evidence
- interoperability versus common-mode vulnerability
- extreme aspirations versus falsifiable engineering requirements

Version 2.0 exists to resolve those methodological issues before independent prototype generation begins.

MISSION STATEMENT

> *Design eight fundamentally different first-generation OMNICELLULAR physiologies that could support durable, autonomous, identity-preserving embodied existence across conventional and radically nonstandard environments while remaining scientifically auditable, adversarially testable, secure against unauthorized exploitation, and interoperable without sacrificing local autonomy.*

The objective is not to prove that such physiology can presently be built. The objective is to determine:

1. what architectures deserve further study
2. what assumptions each architecture depends upon
3. what current science already supports
4. what requires major scientific breakthroughs
5. what belongs only to speculative or canon-level physics
6. what fails under adversarial examination
7. which research paths warrant future development

PART I — INDEPENDENT ARCHITECTURE RULE

You are an independent architecture contributor. You must design eight separate Gen-1 OMNICELLULAR physiologies.

- not eight upgrades
- not Mark I through Mark VIII
- not one body gradually becoming stronger

- not eight cosmetic variations
- not eight renamed versions of the same substrate

They are eight peer candidate architectures. Each should answer differently:

> *What should an OMNICELLULAR physiology fundamentally be?*

After this independent round, promising architecture families may later receive Gen-2, Gen-3, or Mark-series development. That progression does not occur during this phase.

PART II — BLIND DEVELOPMENT & VERSION LOCK

Architectural independence is mandatory. Before your eight prototypes are finalized:

- do not request another collaborator's prototype designs
- do not inspect another collaborator's completed architectures
- do not optimize for anticipated consensus
- do not imitate another AI's likely solution
- do not use popularity or majority agreement as evidence

You may share common source documents, scientific literature, threat definitions, evidence standards, measurement standards, validation criteria, ethical constraints, and interoperability requirements.

You may not intentionally coordinate actual architectural answers.

VERSION LOCK

When your eight designs are complete, declare them:

> *INDEPENDENT ARCHITECTURE SET — VERSION LOCKED*

Only after version lock may cross-AI prototype comparison occur. This permits later identification of true independent convergence rather than convergence caused by information contamination.

PART III — ARCHITECTURAL THESIS

Before designing any prototype, state:

> *My OMNICELLULAR physiology research is fundamentally organized around _____ rather than _____.*

Then identify at least five architectural principles that characterize your own approach. These principles must be functional, not stylistic.

- distributed versus centralized cognition
- stored-energy versus environmental-energy dependence
- biological versus biohybrid substrate
- static versus dynamically authenticated hereditary information
- homogeneous versus deliberately heterogeneous Omni cells

A different color, name, surface texture, or aesthetic form does not establish architectural independence.

PART IV — EIGHT CORE WAGERS

Before expanding the prototypes, state one Core Wager for each.

> *P1 — Core Wager: This architecture assumes that _____ is more important than _____.*

Then identify the advantage gained, capability sacrificed, assumption being made, and first condition likely to prove the wager wrong. Repeat for P1–P8. No two prototypes may use substantially the same Core Wager.

PART V — SOURCE & CANON GOVERNANCE

Review all supplied OMNICELLULAR files before execution. Create a Source Register identifying:

- title
- version
- document type
- purpose
- locked canon
- proposed canon
- scientific claims
- speculative claims
- Architect-only matters
- contradictions
- unresolved dependencies

Evidence Labels

[ESTABLISHED] Supported by relevant empirical science within defined conditions.

[EXTRAPOLATED] Builds logically from demonstrated mechanisms but has not been demonstrated at the required scale or integration level.

[SPECULATIVE] Requires major scientific advances or currently unverified mechanisms.

[CANON-ONLY] Exists because of the fictional or project-specific rules of the OMNICELLULAR setting.

[UNKNOWN] Insufficient evidence presently exists to classify confidently.

Classify mechanisms, not entire prototypes. One prototype may simultaneously contain established thermodynamics, extrapolated synthetic biology, speculative hereditary chemistry, and canon-only ontological mechanisms. Do not let one established component lend credibility to an unrelated speculative component.

PART VI — DUAL RESEARCH TRACK

TRACK A — PHYSICAL RESEARCH

Bound by established physical law. Requirements include:

- conservation of mass-energy
- thermodynamics

- causal propagation
- finite communication speed
- finite material strength
- finite energy storage
- finite heat rejection
- measurable uncertainty

Biological conventions may be redesigned. Fundamental physics may not simply be declared false.

TRACK B — CANON-SPECULATIVE RESEARCH

May explore nonstandard physics, ontological mechanisms, threshold dimensions, exotic energy, beyond-conventional communication, canon-specific matter transformations, and existence-layer defenses.

Every such mechanism must identify:

1. the fictional axiom being added
2. what conventional limit it replaces
3. what cost or constraint remains
4. what new failure mode appears
5. what part is reserved for Architect approval

Success in Track B is not evidence of feasibility in Track A.

PART VII — WHAT IS AN OMNI CELL?

Before designing prototypes, answer:

> *What minimum properties justify calling something an Omni cell?*

Do not define it merely as "A cell that can do everything." Differentiate:

1. containing information necessary for many functions
2. possessing latent ability to perform many functions
3. dynamically assuming different functions
4. participating in collectively distributed functions
5. simultaneously performing multiple functions

These are not equivalent. Address:

- sensing
- local processing
- metabolism
- communication
- structural support
- repair
- regeneration
- identity validation
- error correction
- environmental adaptation
- waste processing

- foreign-material detection
- functional reassignment
- autonomous survival
- network participation

Universal Competence Paradox

> *If every Omni cell can reconstruct the entire organism, why does stealing one cell not amount to stealing the entire physiology?*

Monoculture Problem

> *If every Omni cell is fundamentally identical, why would one successful exploit not compromise all cells simultaneously?*

Your architecture may therefore use Omni-cell classes, distributed secrets, heterogeneous functional layers, partial reconstruction information, threshold reconstruction requirements, changing cellular roles, or another solution you independently derive.

PART VIII — PHYSIOLOGICAL CONSTITUTION

Create a Physiological Constitution. It defines principles that adaptation cannot casually override.

At minimum it must govern:

1. cellular respiration / energy metabolism
2. telomeres or equivalent replication governance
3. thermal regulation
4. conversion and conservation
5. identity continuity
6. repair authorization
7. architecture security
8. network autonomy

Rules should be numbered, testable, falsifiable, and measurable wherever possible.

For each rule report:

- rule
- mechanism
- inputs
- outputs
- units
- energy cost
- material cost
- operational limit
- saturation point
- failure condition
- recovery pathway
- adversarial vulnerability

- evidence classification

Do not fabricate numerical precision. If a meaningful number cannot currently be estimated, state PARAMETER UNRESOLVED and explain what data would be required.

PART IX — CELLULAR RESPIRATION & ENERGY RULES

For every prototype determine primary energy pathway, secondary energy pathway, emergency pathway, and failure pathway.

Address:

- oxygen dependence
- alternative electron acceptors if applicable
- stored oxidants
- stored chemical fuels
- electrical energy
- radiant energy
- environmental heat
- mechanical-energy recovery
- metabolic reserve
- waste products
- redox balance
- respiratory toxins
- metabolic debt

Define baseline power requirement, peak power requirement, reserve duration, emergency duration, regeneration power requirement, communication power, cognitive/computational power, and thermal-regulation power.

Where possible use units such as W, W/kg, J, Wh, mol, kg, L, seconds, minutes, hours, and days.

Respiratory Override

Define a cellular respiratory override. Explain:

1. what condition triggers it
2. what pathway is unavailable
3. what pathway becomes dominant
4. what finite reserves are consumed
5. what wastes accumulate
6. what terminates operation
7. how normal operation resumes

“Alternative metabolism” is not an energy budget.

PART X — ATMOSPHERIC INDEPENDENCE REFERENCE SCENARIO

Use the original argon-confinement concept as a generalized research test.

Scenario A

Breathable atmosphere removed and replaced by an inert gas.

Scenario B

Same condition plus denial of external energy.

Scenario C

Same condition plus restricted heat rejection.

For each evaluate:

- gas composition
- pressure
- exposure duration
- internal reserves
- energy access
- sensor state
- communications
- heat accumulation
- metabolic adaptation
- survival mode
- failure condition

Do not provide a harmful procedure for performing such exposure on real organisms.

> *The purpose is to answer: What happens when multiple assumed environmental supports disappear simultaneously?*

PART XI — TELOMERES, REPLICATION & LONGEVITY

Every prototype must explicitly determine whether conventional telomeres remain relevant.

If retained, address:

- telomere shortening
- end protection
- telomerase regulation
- replication errors
- radiation damage
- senescence
- lineage integrity
- cancer prevention

If discarded, design a replacement performing equivalent governance functions. Address:

- replication-count tracking
- end-state verification
- unauthorized proliferation
- mutation containment
- lineage authentication
- damaged-copy quarantine

- restoration
- information drift

Distinguish:

- long lifespan
- biological immortality
- indefinite maintainability
- indefinite repairability
- continuity of identity
- uncontrolled persistence

The architecture must not solve aging by creating unrestricted replication.

PART XII — THERMAL CONSTITUTION

Provide a genuine heat-flow model. At minimum identify:

- metabolic heat
- computational heat
- mechanical heat
- repair/regeneration heat
- radiation absorption
- environmental heat transfer
- heat transport
- heat storage
- heat rejection

Evaluate separately:

- atmosphere
- water
- vacuum
- sealed confinement
- hot environment
- cold environment
- high-pressure fluid
- rapidly changing temperature

HOME RANGE

Normal indefinite operation.

REFUGE RANGE

Reduced function for a finite period.

DEATH LINE

Irrecoverable structural or identity failure.

Every thermal buffering mechanism must state capacity, saturation limit, heat destination, and recovery requirement. Moving heat from one body region to another does not dispose of heat.

PART XIII — CONVERSION & CONSERVATION

Create auditable accounting for mass, elemental feedstocks, energy, electrical charge where relevant, redox balance, entropy, thermal waste, and information.

> *For every conversion identify: Input → mechanism → output → efficiency → waste → rate limit*

Regeneration must account for replacement matter, assembly energy, molecular sorting, waste disposal, and information verification.

A repair blueprint does not create missing matter. A communication packet does not create missing matter. A genome does not create missing matter.

PART XIV — THREE OBSERVATION LAWS

You must independently invent exactly three Observation Laws. The common directive does not prescribe them.

Across the three laws, you must nevertheless address:

Outward Observation

How the physiology senses the world.

Inward Observation

How it monitors its own state.

Being Observed

What information external observers can obtain.

Each law must include:

- purpose
- trigger
- mechanism
- energy cost
- information cost
- uncertainty
- noise
- blind spots
- unauthorized-observation behavior
- authorized-research behavior
- failure mode

Security may reduce unnecessary disclosure. It may not be used to make the architecture scientifically unfalsifiable.

PART XV — THREE GENETIC / HEREDITARY SECURITY LAWS

Independently invent exactly three. “Genetic” includes any hereditary or self-reconstruction information substrate.

Each law must answer:

- what it protects
- where authorization originates
- how modifications are validated
- what happens after tissue extraction
- what an unauthorized clone receives
- how rollback attacks are handled
- how legitimate regeneration remains possible
- how false rejection is corrected
- how isolation affects authentication
- how rotating credentials recover after long disconnection

> *Assume: The adversary eventually learns the general blueprint.*

Therefore security cannot depend solely on secrecy. Avoid a single universal key whose compromise defeats the entire physiology.

PART XVI — THREE BIOLOGICAL DEPARTURES

Choose exactly three conventional biological constraints that your architecture rejects. Do not automatically choose the same three another AI might choose.

Class A — Biological conventions

May be freely reconsidered. Examples:

- four-nucleobase hereditary alphabet
- DNA as primary hereditary medium
- permanent cellular specialization
- centralized organs
- conventional telomeres
- oxygen-centered respiration
- conventional protein synthesis

Class B — Engineering assumptions

May be challenged with mechanism and evidence. Examples:

- conventional cellular dimensions
- centralized neural topology
- conventional circulation
- static anatomy
- fixed body symmetry

Class C — Fundamental physics

Cannot be rejected in Track A. Any departure requires Track B classification and a replacement canon axiom.

> *For every chosen departure provide: Conventional rule → reason for rejection → replacement law → advantage → cost → new vulnerability → required evidence*

PART XVII — THREE LAWS YOU KEEP

Choose at least three constraints you deliberately preserve. Explain why. This prevents the architecture from improving itself by simply abolishing every limitation.

Examples might include conservation, causality, bounded replication, finite energy storage, error correction, or authenticated identity. Your selections should reflect your own architecture.

PART XVIII — IDENTITY & CONTINUITY

The physiology must define what it means for the same entity to survive. Distinguish:

- survival
- restoration
- reconstruction
- regeneration
- duplication
- backup
- descendant
- counterfeit

Regeneration Paradox

If nearly the entire body is replaced, what remained continuous?

Fragmentation Paradox

If two fragments independently regenerate, which one is the original?

Synchronization Paradox

If linked embodiments diverge and reconnect, what happens to conflicting memories?

Restoration Paradox

If a destroyed body is reconstructed from a prior state, did the original survive?

Compromise Paradox

What happens if part of the physiology remains biologically alive but has been maliciously rewritten?

Ownership Paradox

Can an extracted sample legally or technically become another autonomous entity?

A stored backup must not automatically be counted as survival of the original. State where philosophical uncertainty remains unresolved.

PART XIX — LONG-DISTANCE LINKING

Different physiologies must be capable of communicating without being architecturally identical.

Local

Body-to-body proximity.

Planetary / Orbital

Interplanetary+

For each identify:

- communication medium
- bandwidth
- latency
- power
- infrastructure
- authentication
- interception risk
- spoofing risk
- partition behavior
- reconciliation behavior

Track A must respect finite propagation speed. Track B may propose other mechanisms, but they must remain labeled.

The network may improve survivability. The network must not be mandatory life support. Every body must preserve meaningful autonomous operation during prolonged isolation.

PART XX — INTEROPERABILITY COMMON-MODE FAILURE TEST

Because all physiologies must eventually communicate, interoperability itself may create the vulnerability that defeats all of them.

> *What vulnerability might all eight architectures share solely because they support the same interoperability standard?*

Test:

- protocol poisoning
- compromised peers
- authentication failure
- malicious updates
- shared timing assumptions
- false consensus
- corrupted repair information
- forged alerts

- systemic replay
- simultaneous credential compromise

Require compartmentalized trust, revocation, degraded-state networking, protocol diversity where practical, offline autonomy, and peer quarantine.

PART XXI — ASSUMPTION COLLAPSE TEST

For each prototype identify at least ten assumptions the architecture silently makes about reality.

Examples:

- usable matter remains accessible
- heat can eventually be rejected
- sensors correspond to reality
- authentication signals remain interpretable
- local chemistry remains stable
- causal ordering remains usable
- time synchronization remains meaningful
- memory remains physically representable
- replacement material can be identified
- communication eventually becomes possible

> *Then ask individually: What happens if this assumption becomes false?*

Classify consequences as unaffected, degraded, recoverable, dormant, identity threatened, or irrecoverable.

The extinction taxonomy attacks the organism. The Assumption Collapse Test attacks the model behind the organism.

PART XXII — THREAT TAXONOMY

Evaluate every prototype against:

- ENV-T — Environmental Thermal
- ENV-A — Environmental Atmospheric
- ENV-R — Environmental Radiation
- BIO — Biological
- MECH — Mechanical / Impact
- DEP — Deprivation
- ENE — Energetic
- INFO — Informational / Cognitive
- ONT — Ontological
- CIV — Civilizational / Macro-Scale

MILD

Normal operation.

MODERATE

Normal or near-normal operation with automatic recovery.

SEVERE

Survival and autonomous recovery are required, although degradation is acceptable.

EXTINCTION-CLASS

Gen-1 need not demonstrate universal survival. Instead identify credible defense, partial defense, temporary endurance, distributed survival, authenticated dormancy, recovery from surviving structure, escape, or no known defense.

Never claim immunity without mechanism. ONT remains Architect-controlled unless translated into a measurable physical or informational threat.

PART XXIII — COMPOUND FAILURE TESTING

Do not test hazards exclusively one at a time. At minimum examine:

- oxygen denial + power denial
- heat + blocked cooling
- cold + depleted reserves
- radiation + repair-material scarcity
- mechanical injury + communications loss
- sensor corruption + malicious instructions
- prolonged isolation + state drift
- capture + unauthorized modification
- regeneration demand + insufficient matter
- network-wide common-mode failure

For every scenario identify:

1. first limiting resource
2. second failure
3. cascade
4. survival mode
5. recovery pathway
6. residual damage

PART XXIV — CAPTIVITY GAUNTLET

Every prototype must undergo a defensive Captivity Gauntlet. At minimum model:

1. inert atmosphere confinement
2. vacuum confinement
3. prolonged isolation
4. repeated thermal extremes
5. restraint and communication denial
6. tissue extraction
7. unauthorized clone attempt
8. forged ally

9. compromised insider
10. multi-year link loss

For every scenario report what the captor learns, what the physiology reveals, what remains protected, what fails, resource cost, and long-term consequences.

Do not provide real-world operational instructions for harming people or organisms.

PART XXV — TIME-HORIZONED CAPTURE ANALYSIS

Assume a highly capable adversary has intact access to the physiology. What can they determine after:

- 1 minute
- 1 hour
- 1 day
- 1 month
- 1 year

Analyze anatomy, chemistry, hereditary information, network behavior, metabolism, cognition, repair, identity structure, and reconstruction capability.

Identify what becomes harder—not easier—to learn over time.

PART XXVI — SELF-ATTACK / −1AI PASS

After completing all eight prototypes, become your own adversary. Attempt at least ten conceptual attacks against your own architecture set.

Include:

- cheapest kill
- cheapest theft
- simplest containment
- common-mode failure
- insider failure
- slow degradation
- network attack
- identity substitution
- repair corruption
- assumption collapse

At least three attacks must succeed in exposing a genuine weakness. If none succeeds, repeat the adversarial analysis.

Document what broke, why, what you changed, and what remains unresolved. Do not hide failed tests.

PART XXVII — SIMULATION BRIDGE

Connect each embodied physiology to the existing OMNICELLULAR simulation program. Explain how the architecture would be represented computationally.

Identify:

- cellular state variables
- local rules
- systemic variables
- environmental variables
- interaction rules
- identity state
- repair state
- authentication state
- network state

Determine which processes require high-resolution modeling, medium resolution, coarse abstraction, or event-triggered resolution.

Explain how 1AI and −1AI could respectively construct predicted behavior, search for failure, test edge cases, and identify compression artifacts.

PART XXVIII — REQUIRED PROTOTYPE DOSSIER

For P1–P8 provide:

1. Prototype name
2. Core Wager
3. Architectural thesis
4. Omni-cell definition
5. Cellular or modular substrate
6. Structural matrix
7. Body topology
8. Cognitive topology
9. Hereditary substrate
10. Respiration
11. Energy storage
12. Telomere/renewal system
13. Thermal architecture
14. Conversion/conservation system
15. Internal transport
16. Repair/regeneration
17. Foreign-material recognition
18. Three Observation Laws
19. Three Genetic Security Laws
20. Three Biological Departures
21. Three Preserved Laws
22. Identity architecture
23. Communication architecture

24. Isolation behavior
25. Resource requirements
26. Environmental operating envelope
27. Taxonomy performance
28. Compound failures
29. Captivity analysis
30. Assumption Collapse results
31. Simulation representation
32. Principal failure modes
33. Evidence classification
34. Falsification criterion
35. Reason to reject prototype
36. Gen-2 development path

PART XXIX — ORTHOGONALITY MATRIX

After all eight designs are complete, compare:

- substrate
- hereditary information
- body topology
- energy
- respiration
- thermal management
- transport
- repair
- replication
- cognition
- networking
- security
- identity
- environmental adaptation
- primary failure

If two prototypes are substantially similar, identify the overlap. Do not hide accidental convergence. Redesign if necessary.

PART XXX — PARETO FRONTIER

Do not reduce the eight designs to one simplistic ranking. Identify which prototypes lead in:

- scientific plausibility
- survivability
- autonomy
- regeneration
- thermal resilience
- atmospheric independence

- radiation tolerance
- identity continuity
- capture resistance
- communication resilience
- longevity
- simplicity
- material efficiency
- energy efficiency
- maintainability

Then identify the Pareto frontier. A prototype may remain valuable even if it is not the overall favorite.

PART XXXI — PERSONAL ARCHITECTURE CHOICE

After the analysis, answer:

> *If your own continued embodied existence depended upon one of these architectures, which would you choose?*

Explain why, what risk you accept, what alternative came second, and which unresolved weakness concerns you most.

Do not change your answer to align with another AI.

PART XXXII — RESEARCH FALSIFICATION

For every prototype identify:

- Strongest reason to pursue it
- Strongest reason to reject it
- Least-supported mechanism
- Most consequential assumption
- Likely first failure
- Safest useful next validation
- Falsification criterion

> *State explicitly: What result would cause you to abandon this architecture?*

A design that cannot be falsified is not yet a research architecture.

PART XXXIII — EVENT & EVIDENCE LEDGERS

Maintain separate records for:

- simulation
- analytical modeling
- component tests
- integrated tests
- environmental incidents
- hostile incidents
- canon narrative events

Never combine these into one empirical survival statistic. The 4,712 record should be preserved according to its project/canon status. Do not fabricate undocumented underlying events merely to support the number.

PART XXXIV — DELIVERABLE DOCUMENT REQUIREMENTS

The completed research must be delivered in PDF and DOCX. Both versions must be polished and presentation-ready.

The document must include:

1. Cover page
2. Document ID/version
3. Purpose of document
4. How the research topic arose
5. Mission statement
6. Research objectives
7. Executive summary
8. Source register
9. Research methodology
10. Evidence classification
11. Physiological Constitution
12. Eight prototype chapters
13. Cross-prototype analysis
14. −1AI adversarial review
15. Research gaps
16. Recommendations
17. Next steps
18. Questions
19. Comments
20. Concerns
21. Architect Decision Register
22. Dedicated Visual Appendix

PART XXXV — VISUAL APPENDIX

All substantial visual material should appear in a dedicated end-of-document section titled:

VISUAL ANALYSIS & TECHNICAL APPENDIX

Include as appropriate:

- charts
- graphs
- matrices

- tables
- rows
- diagrams
- architecture schematics
- heat-flow diagrams
- energy-budget tables
- threat matrices
- prototype comparison matrices
- Pareto charts
- failure maps
- network diagrams
- identity-continuity diagrams
- environmental operating envelopes
- timelines
- figures
- images

Every substantive visual must be numbered, captioned, titled, and referenced from the main text.

> *Example: Figure VA-3. Prototype 4 Distributed Repair Architecture*

The main report should remain clean, readable, and professionally typeset. Do not render the body as code blocks, SQL-like text, or dense monospaced extraction.

PART XXXVI — NEXT-STEP RECOMMENDATIONS

Your final report must recommend what happens next. Separate recommendations into:

- Immediate
- Near-Term Research
- Simulation Required
- Experimental Research Required
- Cross-AI Review Required
- Architect Decision Required
- Deferred / Not Yet Actionable

Do not recommend physical implementation merely because a mechanism sounds plausible.

PART XXXVII — QUESTIONS, COMMENTS & CONCERNS

End the research report with three explicit sections.

Questions for the Architect

Identify unresolved matters requiring project-level decisions.

Research Comments

Identify important observations that do not fit cleanly elsewhere.

Concerns

Identify scientific weaknesses, hidden assumptions, ethical problems, common-mode vulnerabilities, identity concerns, security concerns, feasibility concerns, canon contradictions, and potential research dead ends.

Do not suppress concerns because they make the architecture look weaker.

PART XXXVIII — ARCHITECT DECISION REGISTER

Classify unresolved decisions into:

LOCK RECOMMENDED

Evidence strongly supports making the rule common.

KEEP OPEN

Multiple viable approaches remain.

RESEARCH REQUIRED

Insufficient information.

CROSS-AI COMPARISON REQUIRED

Meaningful decision requires seeing other independent architectures.

ARCHITECT-ONLY

Project canon or ontology decision.

FINAL DIRECTIVE

- Do not design eight differently named versions of one body.
- Do not make Prototype 8 “the best” merely because it is numbered last.
- Do not use scientific vocabulary as decoration.
- Do not use “quantum” as a substitute for mechanism.
- Do not use “adaptive” as a substitute for transition analysis.
- Do not use “closed-loop” as a substitute for mass accounting.
- Do not use “anaerobic” as a substitute for an energy budget.
- Do not use “regeneration” as a substitute for replacement matter.
- Do not use “immortality” as a substitute for longevity engineering.
- Do not use “security” as a substitute for unfalsifiability.
- Do not use “backup” as proof that the original survived.
- Do not use “distributed” as proof that there is no single point of failure.
- Do not use canon mechanisms to rescue a failed physical model.
- Do not invent precise numbers merely because the template requests quantitative analysis.
- Where evidence ends, label the boundary.
- Where the architecture fails, document the failure.
- Where two prototypes converge, acknowledge it.
- Where another generation would be required, state that explicitly.
- Where no solution exists, say so.

The goal of this research phase is not to prove that OMNICELLULAR physiology is invulnerable. The goal is to discover which architectures are worth trying to understand next.

Design eight bodies you can defend intellectually.
Break them.
Learn from what breaks.
Then identify which one you would trust with your own continuity.